



Categories and Classifying Spaces

Connor Porter

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School of Mathematics,
Cardiff University

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Executive Summary

This dissertation investigates the link between algebra and topology through the idea of classifying spaces. Given a discrete group G , we can construct a topological space BG that reflects the algebraic structure of the group. A fundamental theorem states that

$$H_n(G; \mathbb{Z}) \cong H_n(BG; \mathbb{Z}), \quad \text{for all } n \geq 0,$$

which says that the homology of a group G is isomorphic to the homology of its classifying space, with integer coefficients.

The project is organised in three parts. I begin by setting up the background from category theory and simplicial sets, which naturally leads to the construction of classifying spaces. A main contribution to this part is the proof that the geometric realisation of the nerve of a group comes with a natural Δ -complex structure. This proof is often taken for granted in textbooks, but here it is written in full, making the connection between algebra and geometry completely clear.

The second part turns to homology and cohomology. I explain how simplicial and chain complexes are built, how their (co)homology groups are defined, and how the *Universal Coefficient Theorem* links homology and cohomology. Worked examples, such as the circle S^1 , demonstrate how the theory works in practice and serve as a check against previous calculations.

The final part brings the algebraic and topological approaches together. On the topological side, we compute using the simplicial chain complex of the classifying space BG . On the algebraic side, we compute the homology groups directly using a free $\mathbb{Z}G$ -resolution of $\mathbb{Z}$. By showing that these constructions agree, I prove the central theorem. To highlight the usefulness of the agreement, I compute the homology of $B\mathbb{Z}/n$ in both ways and find that they are equivalent.

Overall, this dissertation shows that approaching algebra through category theory can lead to the same conclusions as studying topology directly. The results highlight how much can be achieved with fairly elementary tools, while also pointing toward natural extensions to the theory, such as cohomology rings or the geometry of more complicated classifying spaces. More broadly, the work highlights how classifying spaces serve as one of the many connections between algebra and topology, so that developments in one area can improve results in the other.

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Introduction

One of the most common questions in modern mathematical research is how do different areas interact? Take groups and topology, for instance. Groups are algebraic structures that describe symmetry and transformations. Topology, on the other hand, focuses on the properties of spaces that depend only on shape and continuous deformation. These subjects seem distinct, but there's a natural question we can ask: *can we use topological methods to study groups, and can algebraic invariants of groups tell us something about topology?* What at first appear to be two separate subjects turn out to fit together more naturally than one might expect.

This is exactly what classifying spaces do. Given any discrete group G , we can construct a topological space BG that turns the algebraic structure of G into homotopy-theoretic information. The remarkable thing is that this translation works both ways: we can also recover the algebraic properties of G by studying the topology of BG .

Why should we care about this? Classifying spaces appear in many areas of mathematics:

- In algebraic topology, they provide examples of Eilenberg–MacLane spaces $K(G, 1)$ (Rotman 1988), which are fundamental building blocks in homotopy theory.
- In geometry, they classify principal G –bundles: to understand all such bundles over a space X , it is enough to study the maps $X \rightarrow BG$ up to homotopy.
- In algebra, they offer a shortcut: instead of computing the homology of a group directly via algebraic methods, we can compute the homology of its classifying space using topological methods.

The heart of this project is to prove that these two approaches, algebraic and topological, give the same results. Specifically, for any discrete group G :

$$H_n(G; \mathbb{Z}) \cong H_n(BG; \mathbb{Z}).$$

This theorem (see Corollary 4.1.25 in Löh 2019) tells us that the homology of the group G (computed algebraically) is the same as the homology of its classification space (computed topologically).

My goal in this dissertation is to highlight this connection. I begin by developing the necessary background from category theory and simplicial sets, which are the tools needed to construct classifying spaces. I then show how to compute these spaces using two different methods: the topological approach (via simplicial complexes) and the algebraic approach (via free resolutions). The central result comes when we prove that these methods yield identical results.

Before this, I also include an account of cohomology and *Universal Coefficient Theorem*. This provides an algebraic link between homology and cohomology, showing how the latter can be recovered from the former together with Hom and Ext terms. By working through examples such as the circle S^1 , I demonstrate how and when the theorem works and confirm the consistency of examples with earlier homology computations.

Finally, I apply the theory to explicit examples with cyclic groups such as $\mathbb{Z}$ and $\mathbb{Z}/n$, demonstrating how the main theorem can be applied. By the end, I aim to show

how ideas from category theory, algebra, and topology come together in the theory of classifying spaces, and how cohomology provides an additional perspective on the material.

1 Introduction to Category Theory

Category theory provides a unifying language for modern mathematics and it abstracts the idea of relationships between mathematical structures. Rather than focussing solely on the internal details of specific objects, category theory mostly emphasises the morphisms between the objects.

This perspective enables diverse areas of mathematics to be described in a common way: sets, groups, topological spaces, and vector spaces, among others, can all be viewed as categories. Functors, which are mappings between categories, then allow us to study entire mathematical areas as objects in their own right.

In the context of topology and algebra, category theory plays a fundamental role in defining topological constructions from algebraic and combinatorial information, such as simplicial sets and classifying spaces. To understand classifying spaces, it is essential to begin by defining the basics: categories, functors, and natural transformations.

This section directly follows the Basic Category Theory lecture notes by Leinster (2016), using his chapter on categories, functors, and natural transformations as a base for the start of this project.

1.1 Categories

Definition 1.1 (Category). A *category* $\mathcal{C}$ consists of:

- A class of objects, denoted $\text{ob}(\mathcal{C})$.
- For every pair of objects $A, B \in \text{ob}(\mathcal{C})$, a set of morphisms (sometimes called maps or arrows) $\text{Hom}_{\mathcal{C}}(A, B)$.
- A binary operation called *composition*: for morphisms $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $g \in \text{Hom}_{\mathcal{C}}(B, C)$, there exists a morphism $g \circ f \in \text{Hom}_{\mathcal{C}}(A, C)$.
- For every object $A \in \text{ob}(\mathcal{C})$, there exists an identity morphism $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$.

This structure must satisfy the following axioms:

1. **Associativity:** For all $f \in \text{Hom}_{\mathcal{C}}(A, B)$, $g \in \text{Hom}_{\mathcal{C}}(B, C)$, and $h \in \text{Hom}_{\mathcal{C}}(C, D)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

2. **Identity:** For every $f \in \text{Hom}_{\mathcal{C}}(A, B)$,

$$\text{id}_B \circ f = f = f \circ \text{id}_A.$$

Example 1.2 (The category **Set**). One common example of a category that will show up throughout this project is the category of sets, denoted **Set**. It is defined as follows:

- **Objects:** The objects of **Set** are all sets.
- **Morphisms:** The morphisms are maps between sets. That is, a morphism $f : A \rightarrow B$ in **Set** is simply a map from the set A to the set B .

- **Composition:** Given functions $f : A \rightarrow B$ and $g : B \rightarrow C$, their composition is the usual composition of maps:

$$g \circ f : A \rightarrow C, \quad (g \circ f)(a) = g(f(a)).$$

- **Identity morphisms:** For each set A , the identity morphism $\text{id}_A : A \rightarrow A$ is the identity map, defined by $\text{id}_A(a) = a$ for all $a \in A$.

The category **Set** satisfies the axioms of a category:

- Map composition is associative.
- Identity maps for sets act as neutral elements for composition.

Example 1.3 (The category **Top**). The category **Top** consists of topological spaces and continuous maps between them:

- **Objects:** The objects are all topological spaces, i.e. each object is a set equipped with a topology.
- **Morphisms:** The morphisms are continuous functions between topological spaces: A morphism $f : X \rightarrow Y$ is continuous if for every open set $U \subseteq Y$, the preimage $f^{-1}(U) \subseteq X$ is open.
- **Composition:** Composition of morphisms is the normal composition of functions, which preserves their continuity.
- **Identity morphisms:** For each space X , the identity function $\text{id}_X : X \rightarrow X$ is continuous and serves as the identity morphism.

The category **Top** satisfies the axioms of a category:

- Continuous function composition is associative.
- Identity functions for topological spaces act as neutral elements for composition.

1.2 Functors

Definition 1.4 (Functor). A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$ consists of:

- A function $F : \text{ob}(\mathcal{C}) \rightarrow \text{ob}(\mathcal{D})$.
- For each pair of objects $A, B \in \text{ob}(\mathcal{C})$, a function

$$F_{A,B} : \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B)),$$

often written as $F(f)$ for a morphism $f : A \rightarrow B$.

These assignments must preserve the structure of a category:

1. **Preservation of identities:** $F(\text{id}_A) = \text{id}_{F(A)}$.

2. **Preservation of composition:** For $f : A \rightarrow B$ and $g : B \rightarrow C$ in $\mathcal{C}$,

$$F(g \circ f) = F(g) \circ F(f).$$

Remark 1.5. Functors come in two main types: *covariant* and *contravariant*. A covariant functor preserves the direction of morphisms: a map $f : A \rightarrow B$ in the starting category is sent to a map $F(f) : F(A) \rightarrow F(B)$ in the target category. In contrast, a contravariant functor reverses arrows: the same morphism $f : A \rightarrow B$ is sent to a map $F(f) : F(B) \rightarrow F(A)$. Both kinds preserve identities, but while covariant functors preserve composition, contravariant functors reverse the order of composition. This distinction is important to keep in mind, especially in Chapter 2.

Example 1.6 (The forgetful functor from **Top** to **Set**). There is a functor $F : \mathbf{Top} \rightarrow \mathbf{Set}$ that *forgets the topology*, assigning to each topological space its underlying set.

- **On objects:** For a topological space (X, τ) , define $F(X) = X$, the underlying set of points, ignoring the topology.
- **On morphisms:** For a continuous function $f : X \rightarrow Y$, define $F(f) = f$, now regarded simply as a map between sets.

This defines a functor because:

- It preserves identity morphisms:

$$F(\text{id}_X) = \text{id}_X = \text{id}_{F(X)}.$$

- It preserves composition:

$$F(g \circ f) = g \circ f = F(g) \circ F(f),$$

for any composable continuous maps $f : X \rightarrow Y$, $g : Y \rightarrow Z$.

This functor is an example of a *forgetful functor*, because it forgets the topological structure and retains only the underlying set. Forgetful functors also exist for many other categories such as **Grp**, the category of groups, and **Mod**(R), the category of R -modules, for example.

Example 1.7 (The doubling functor on **Set**). Define a functor $F : \mathbf{Set} \rightarrow \mathbf{Set}$ as follows:

- **On objects:** For any set A , let $F(A) = A \times A$, the Cartesian product of A with itself.
- **On morphisms:** For any function $f : A \rightarrow B$, define $F(f) : A \times A \rightarrow B \times B$ by

$$F(f)(a_1, a_2) = (f(a_1), f(a_2)).$$

This defines a functor because:

- It preserves identities: for every set A ,

$$F(\text{id}_A)(a_1, a_2) = (\text{id}_A(a_1), \text{id}_A(a_2)) = (a_1, a_2) = \text{id}_{A \times A}(a_1, a_2).$$

- It preserves composition: for functions $f : A \rightarrow B$ and $g : B \rightarrow C$,

$$\begin{aligned} F(g \circ f)(a_1, a_2) &= ((g \circ f)(a_1), (g \circ f)(a_2)) \\ &= (g(f(a_1)), g(f(a_2))) \\ &= F(g)(F(f)(a_1, a_2)). \end{aligned}$$

$$F(g \circ f) = F(g) \circ F(f).$$

1.3 Natural Transformations

Definition 1.8 (Natural Transformation). Given two functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$, a *natural transformation* $\eta : F \Rightarrow G$ assigns to each object $A \in \text{Ob}(\mathcal{C})$ a morphism $\eta_A : F(A) \rightarrow G(A)$ in $\mathcal{D}$, such that for every morphism $f : A \rightarrow B$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \eta_A \downarrow & & \downarrow \eta_B \\ G(A) & \xrightarrow{G(f)} & G(B) \end{array}$$

That is,

$$\eta_B \circ F(f) = G(f) \circ \eta_A$$

for all $f : A \rightarrow B$ in $\mathcal{C}$. The arrow $\eta : F \Rightarrow G$ indicates that η is a natural transformation between the functors F and G .

Example 1.9. In this example, we define a natural transformation from the identity functor to the doubling functor.

Let $\text{id}_{\mathbf{Set}}$ denote the identity functor on $\mathbf{Set}$, and let $F : \mathbf{Set} \rightarrow \mathbf{Set}$ be the doubling functor defined in the previous subsection. Define a natural transformation $\eta : \text{id}_{\mathbf{Set}} \Rightarrow F$ as follows:

For each set A , define

$$\eta_A : A \rightarrow A \times A, \quad \eta_A(a) = (a, a).$$

(This morphism has a name; the *diagonal map*.)

We now verify that this defines a natural transformation. For every function $f : A \rightarrow B$, the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\eta_A} & A \times A \\ f \downarrow & & \downarrow f \times f \\ B & \xrightarrow{\eta_B} & B \times B \end{array}$$

Indeed, for all $a \in A$, we have:

$$(f \times f)(\eta_A(a)) = (f \times f)(a, a) = (f(a), f(a)) = \eta_B(f(a)).$$

Therefore,

$$(f \times f) \circ \eta_A = \eta_B \circ f,$$

so the diagram commutes, and η is a natural transformation from $\text{id}_{\mathbf{Set}}$ to F .

2 Simplicial Sets

A simplicial set is a way of describing a space using simplices (like points, edges and triangles etc.), but written in a purely combinatorial way. They play a central role in modern algebraic topology, especially in homotopy theory, classifying spaces, and higher category theory.

Formally, a simplicial set is a contravariant functor from the simplex category Δ to the category of sets. Intuitively, this means it organises vertices, edges, triangles, and higher simplices, along with the face and degeneracy maps that record how these simplices fit together.

In this section, we introduce the simplex category Δ , define simplicial sets precisely, and describe the associated face and degeneracy maps. Our explanation follows in the style of, R uschoff (2017) (specifically Chapters 2.3 and 2.4), whose presentation and definitions of simplicial sets and their geometric realisation we adopt here.

2.1 The Simplex Category Δ

Definition 2.1 (Simplex Category). The *simplex category*, denoted Δ , is the category defined as follows:

- **Objects** are the finite ordered sets

$$[n] := \{0 < 1 < \dots < n\}$$

for each integer $n \geq 0$.

- **Morphisms** are the order-preserving functions

$$\phi : [m] \rightarrow [n], \quad \text{i.e.} \quad i \leq j \implies \phi(i) \leq \phi(j), \quad \text{for all } i, j \in [m]$$

That is, a morphism in Δ is a map that respects the ordering of the integers in the sets. This category retains the combinatorics of simplices and is fundamental to the definition of simplicial sets.

More specifically, every morphism in Δ can be expressed as a composition of maps from the following two families:

- *Coface maps* $\delta^i : [n-1] \rightarrow [n]$, which are injective order-preserving maps that omit the i -th element in their image;
- *Codegeneracy maps* $\sigma^i : [n+1] \rightarrow [n]$, which are surjective order-preserving maps that repeat the i -th element of their domain.

2.2 Simplicial Sets

Remark 2.2. Given a category $\mathcal{C}$, its *opposite category*, denoted $\mathcal{C}^{\text{op}}$, is defined as follows:

- It has the same objects as $\mathcal{C}$;
- For each morphism $f : A \rightarrow B$ in $\mathcal{C}$, there is a corresponding morphism $f^{\text{op}} : B \rightarrow A$ in $\mathcal{C}^{\text{op}}$;

- Composition is defined by $f^{\text{op}} \circ g^{\text{op}} := (g \circ f)^{\text{op}}$, reversing the order of composition from $\mathcal{C}$.

The identity morphisms of course remain unchanged under this operation.

Definition 2.3 (Simplicial Set). A *simplicial set* is a functor

$$X : \Delta^{\text{op}} \rightarrow \mathbf{Set},$$

where Δ is the simplex category and $\mathbf{Set}$ is the category of sets.

That is, a simplicial set assigns to each object $[n] \in \Delta$ a set $X_n := X([n])$, whose elements are called *n-simplices*, and to each morphism $\phi : [m] \rightarrow [n]$ in Δ , a function

$$X(\phi) : X_n \rightarrow X_m$$

in the opposite direction. In particular, when ϕ is one of the coface or codegeneracy maps in Δ , the induced functions $X(\phi)$ are called the *face* and *degeneracy* maps of the simplicial set.

- The sets $X_0, X_1, X_2, \dots$ contain, respectively, the 0-simplices (vertices), 1-simplices (edges), 2-simplices (triangles), and so on.
- The structure maps include:
 - *Face maps* $d_i = X(\delta^i) : X_n \rightarrow X_{n-1}$, corresponding to the omission of the i -th vertex of an n -simplex;
 - *Degeneracy maps* $s_i = X(\sigma^i) : X_n \rightarrow X_{n+1}$, corresponding to the repetition of the i -th vertex.

2.3 Geometric Realisation

Although simplicial sets are combinatorial, they are often used to model topological spaces. The construction that assigns a topological space to a simplicial set is called geometric realisation. It constructs a space by connecting standard topological simplices in a way that reflects the face and degeneracy structure within the simplicial set. This provides us with a bridge between discrete and continuous information, enabling the geometric interpretation of combinatorial data.

Definition 2.4 (Geometric Realisation). Let $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ be a simplicial set. The *geometric realisation* of X , denoted $|X|$, is the topological space defined as the quotient

$$|X| := ({}_{n \geq 0} X_n \times |\Delta^n|) / \sim,$$

where $|\Delta^n| \subset \mathbb{R}^{n+1}$ is the *standard topological n-simplex*:

$$|\Delta^n| := \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_{i=0}^n t_i = 1 \right\}.$$

The equivalence relation $\sim$ identifies points according to the simplicial structure: for each morphism $\phi : [m] \rightarrow [n]$ in Δ , the simplex $x \in X_n$, and point $t \in |\Delta^m|$, we impose the relation

$$(X(\phi)(x), t) \sim (x, \phi_*(t)),$$

where $\phi_* : |\Delta^m| \rightarrow |\Delta^n|$ is the affine map corresponding to ϕ . Specifically, each point of $|\Delta^m|$ can be written uniquely as

$$t_0 e_0 + t_1 e_1 + \cdots + t_m e_m, \quad t_i \geq 0, \quad \sum_{i=0}^m t_i = 1,$$

where e_i denotes the i -th vertex of the standard simplex. The map ϕ_* is then defined by

$$\phi_* \left(\sum_{i=0}^m t_i e_i \right) = \sum_{i=0}^m t_i e_{\phi(i)}.$$

In other words, ϕ_* sends vertices of $|\Delta^m|$ to vertices of $|\Delta^n|$ according to ϕ , and extends linearly across the whole simplex.

The geometric realisation $|X|$ is constructed by taking one copy of the standard topological n -simplex $|\Delta^n|$ for each n -simplex in X_n , and then glueing these simplices together according to the face and degeneracy maps defined by the functor X . The resulting topological space then contains the combinatorial structure of the simplicial set.

We now show how this construction works via explicit examples for small values of n .

Example 2.5 ($n = 0$). Let X be a simplicial set such that $X_0 = \{x\}$ and all higher simplices are degenerate. This is the constant simplicial set on a point. Its geometric realisation is a single point:

$$|X| \cong \{*\}$$

Example 2.6 ($n = 1$). Let X be the simplicial set $X = \text{Hom}_\Delta(-, [1])$, where $[1] = \{0 < 1\}$ is the ordinal in the simplex category. Then:

- $X_0 = \text{Hom}_\Delta([0], [1]) = \{0, 1\}$, the two order-preserving maps selecting the vertices 0 and 1.
- $X_1 = \text{Hom}_\Delta([1], [1])$ has three elements: the identity map $\text{id}_{[1]}$, and the two constant maps $c_0 : [1] \rightarrow [1]$ (sending everything to 0) and $c_1 : [1] \rightarrow [1]$ (sending everything to 1). The only non-degenerate 1-simplex is $\text{id}_{[1]}$, while c_0 and c_1 are degenerate since they arise from applying the degeneracy map s_0 to the vertices 0 and 1.

Hence, its geometric realisation is the closed interval:

$$|X| \cong [0, 1].$$

Example 2.7 ($n = 2$). Let X be the simplicial set $X = \text{Hom}_\Delta(-, [2])$, where $[2] = \{0 < 1 < 2\}$ is the ordinal in the simplex category. Then:

- $X_0 = \text{Hom}_\Delta([0], [2]) = \{0, 1, 2\}$, the three order-preserving maps selecting the vertices 0, 1, and 2.
- $X_1 = \text{Hom}_\Delta([1], [2])$ has six elements: three non-degenerate maps ($0 \mapsto i, 1 \mapsto j$) for $i < j$, corresponding to the edges $0-1$, $1-2$, and $0-2$; and three degenerate maps, the constant maps sending everything to a single vertex 0, 1, or 2.

- $X_2 = \text{Hom}_\Delta([2], [2])$ has several elements. The unique non-degenerate 2-simplex is the identity map $\text{id}_{[2]}$, while the others are degenerate, arising from collapsing vertices via surjections $[2] \rightarrow [1]$ or $[2] \rightarrow [0]$.

Hence, its geometric realisation is a filled-in triangle:

$$|X| \cong |\Delta^2|.$$

Remark 2.8. In general, for the representable simplicial set $\Delta[n] = \text{Hom}_\Delta(-, [n])$, the non-degenerate simplices are exactly the injective order-preserving maps $[k] \rightarrow [n]$, (see Goerss and Jardine 1999, Chapter 3.4). Thus, $\Delta[n]$ has $\binom{n+1}{k+1}$ non-degenerate k -simplices, corresponding to the $(k+1)$ -element subsets of $\{0, 1, \dots, n\}$. Its geometric realisation $|\Delta[n]|$ is the standard topological n -simplex.

3 The Nerve of a Category

The nerve construction gives us a method to assign any small category with a simplicial set, in which n -simplices correspond to “chains” of n composable morphisms. This allows objects and composable morphisms of a category to be represented combinatorially, in a format that can then be realised as a topological space using geometric realisation. In particular, the nerve contains information of how morphisms in a category compose, representing them as simplices.

The definition of the nerve involves identifying some object of Δ as a category, so we must cover that first. We proceed using knowledge from Madsen and Patchkoria (2017), namely Example 1.5, and Definitions 1.6 and 1.7.

Example 3.1. Consider an object $[n]$ in the simplex category Δ . Recall that $[n] := \{0 < 1 < \dots < n\}$ for some non-negative integer n . We may view this ordered set as a category by taking the elements $\{0, 1, \dots, n\}$ as the objects, and declaring that there is a unique morphism $\alpha : i \rightarrow j$ whenever $i \leq j$. In particular, each i has an identity morphism $i \rightarrow i$, and for $i < j$ we get a single non-identity arrow. In its simplest form, this category $[n]$ can be displayed as

$$0 \longrightarrow 1 \longrightarrow \dots \longrightarrow n.$$

Remark 3.2. A category is said to be *small* if both its collection of objects and its collection of morphisms form sets (rather than classes). This condition ensures that constructions such as the nerve, which involve forming sets of functors or simplices, are well-defined within standard set theory.

Definition 3.3 (Nerve of a Category). Let $\mathcal{C}$ be a small category. The *nerve* of $\mathcal{C}$, denoted $N\mathcal{C}$, is the simplicial set

$$N\mathcal{C} : \Delta^{\text{op}} \rightarrow \mathbf{Set}$$

defined as follows:

- For each $[n] \in \Delta$, let $N\mathcal{C}_n := \text{Fun}([n], \mathcal{C})$, the set of functors from the ordinal category $[n]$ to $\mathcal{C}$. An element of $N\mathcal{C}_n$ is a string of n composable morphisms in $\mathcal{C}$:

$$A_0 \xrightarrow{f_1} A_1 \xrightarrow{f_2} \dots \xrightarrow{f_n} A_n.$$

- For a morphism $\phi : [m] \rightarrow [n]$ in Δ , the map

$$N\mathcal{C}(\phi) : N\mathcal{C}_n \rightarrow N\mathcal{C}_m$$

is defined by precomposition: $F \mapsto F \circ \phi$.

This construction is functorial in $\mathcal{C}$, meaning that a functor $F : \mathcal{C} \rightarrow \mathcal{D}$ induces a map $N(F) : N\mathcal{C} \rightarrow N\mathcal{D}$ of simplicial sets. Once we apply geometric realisation, we obtain a topological space $|N\mathcal{C}|$, called the *classifying space* of $\mathcal{C}$, denoted $B\mathcal{C}$.

We now see this construction in action through a small example and relate it to the case where a group is viewed as a one-object category.

Example 3.4. First, let us begin by working through an example using a small category. Let $\mathcal{C}$ be the category such that:

- Objects: A and B
- Morphisms: $id_A, id_B, f : A \rightarrow B$

There are no other morphisms, so f does not compose with any other non-trivial morphisms. The structure of the nerve $N\mathcal{C}$ can be described as follows:

- $N\mathcal{C}_0 = \{A, B\}$: the 0-simplices correspond to the objects of $\mathcal{C}$.
- $N\mathcal{C}_1 = \{id_A, id_B, f\}$: the 1-simplices correspond to the morphisms of $\mathcal{C}$.
- $N\mathcal{C}_2$ consists of composable pairs of morphisms. Since the only non-identity morphism is $f : A \rightarrow B$, the non-degenerate 2-simplices come from compositions like:

$$A \xrightarrow{id_A} A \xrightarrow{f} B, \quad A \xrightarrow{f} B \xrightarrow{id_B} B.$$

In this example, there are no 2-simplices involving two non-identity morphisms, because there are no such composable pairs. This implies that all of the 2-simplices are degenerate.

Example 3.5 (The nerve of a group). Let G be a group. We can view G as a category $\mathcal{C}_G$ with:

- A single object $*$,
- A morphism $g : * \rightarrow *$ for each element $g \in G$,
- Composition given by group's binary operation: $g \circ h := gh$.

This category has exactly one object, and its morphisms form a group under composition. The nerve $N\mathcal{C}_G$ then yields a simplicial set.

- $N\mathcal{C}_{G0} = \{*\}$: there is only one 0-simplex, corresponding to the single object.
- $N\mathcal{C}_{G1} = G$: the 1-simplices are the morphisms $* \rightarrow *$, i.e., all the group elements.
- $N\mathcal{C}_{G2} = G \times G$: each 2-simplex corresponds to a composable pair of morphisms $g_1, g_2 \in G$, forming the diagram:

$$* \xrightarrow{g_1} * \xrightarrow{g_2} *.$$

- In general, $N\mathcal{C}_{Gn} = G^n$: an n -simplex is a string of n composable morphisms (group elements), representing a functor $[n] \rightarrow \mathcal{C}_G$.

The face and degeneracy maps in this simplicial set are defined in terms of group composition and identity:

- Face maps $d_i : G^n \rightarrow G^{n-1}$ correspond to omitting the i -th vertex, so either it drops the first or last group element (if $i = 0$ or $i = n$), or composes the adjacent ones (else).
- Degeneracy maps $s_i : G^n \rightarrow G^{n+1}$ correspond to repeating the i -th vertex, so an extra element is required to map between the i -th vertex and itself. The only way to do this is to use the identity morphism, i.e. in this case, the identity element $e \in G$.

Then, the geometric realisation $|N\mathcal{C}_G|$ is called the *classifying space* of G , denoted BG .

4 Simplicial Homology and Chain Complexes

Simplicial homology provides a way of extracting algebraic invariants from spaces by breaking them down into simple building blocks. In terms of nerves and classifying spaces, these invariants allow us to study categories and groups using methods from algebraic topology. To do this, we first study simplices and Δ -complexes, which gives a much more flexible method for constructing spaces from simplices than standard methods.

Much of the presentation here follows Section 2.1 in Hatcher's *Algebraic Topology* (Hatcher 2002), where simplicial homology is developed using chain complexes and boundary maps. We adapt his approach, introducing chain groups and boundary maps in this setting before turning to the resulting simplicial homology groups.

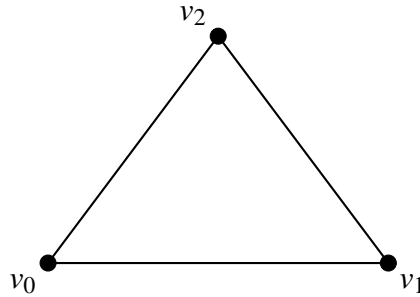
4.1 Simplicial Complexes and Δ -Complexes

Definition 4.1 (Boundary of an n -simplex). Let $[v_0, v_1, \dots, v_n]$ be an n -simplex with vertices $v_0, v_1, \dots, v_n$. The *boundary* of $[v_0, v_1, \dots, v_n]$, denoted $\partial[v_0, v_1, \dots, v_n]$, is the union of its $(n-1)$ -dimensional faces:

$$\partial[v_0, v_1, \dots, v_n] = \bigcup_{i=0}^n [v_0, \dots, \check{v}_i, \dots, v_n],$$

where the notation $\check{v}_i$ indicates that the vertex v_i is omitted.

Example 4.2 (Boundary of a 2-simplex). A 2-simplex $[v_0, v_1, v_2]$ can be represented as a triangle:



Its boundary is the union of its three edges:

$$\partial[v_0, v_1, v_2] = [v_0, v_1] \cup [v_1, v_2] \cup [v_0, v_2].$$

Definition 4.3 (Open n -simplex). Let Δ^n denote the standard n -simplex. Its boundary, denoted $\partial\Delta^n$, is the union of all faces of Δ^n . The *open n -simplex*, denoted $\mathring{\Delta}^n$, is defined as the interior of Δ^n , i.e.

$$\mathring{\Delta}^n := \Delta^n \setminus \partial\Delta^n.$$

Example 4.4 (Open 2-simplex). A 2-simplex $[v_0, v_1, v_2]$ is represented geometrically by a filled triangle. Its boundary consists of the three edges $[v_0, v_1]$, $[v_1, v_2]$, and $[v_0, v_2]$. Removing this boundary yields the open 2-simplex, which is just the interior of the triangle with all the edges and vertices deleted. Algebraically, this can be written as

$$[v_0, v_1, v_2] \setminus ([v_0, v_1] \cup [v_1, v_2] \cup [v_0, v_2]).$$

Turning an n -simplex into its open version is the same for any value of n , so we shall just move on to the definition of a Δ -complex, (Hatcher 2002, Chapter 2.1):

Definition 4.5 (Δ -complex). A Δ -complex on a topological space X consists of a collection of maps

$$\sigma_\alpha : \Delta^n \rightarrow X,$$

called *characteristic maps*, indexed by α , where each σ_α maps the standard topological n -simplex into X , satisfying the following conditions:

1. The restriction $\sigma_\alpha|_{\mathring{\Delta}^n}$ to the interior of the simplex is injective, and the images of these interiors form a partition of X . That is, every point in X lies in the image of exactly one such interior map.
2. Each face of a simplex Δ^n is mapped into X via a composition of a coface map $\delta^i : \Delta^{n-1} \rightarrow \Delta^n$ and another characteristic map σ_β . That is, for each face, there exists β such that

$$\sigma_\alpha \circ \delta^i = \sigma_\beta.$$

3. A subset $A \subseteq X$ is open if and only if each preimage $\sigma_\alpha^{-1}(A) \subseteq \Delta^n$ is open in the standard topology on the simplex.

A Δ -complex is a generalisation of a simplicial complex that allows for a more flexible way of “glueing” simplices together. In a simplicial complex, each simplex is uniquely determined by a set of vertices, and the intersection of any two simplices is either empty or a common face of both. A Δ -complex relaxes the conditions, i.e. simplices may share faces in more general ways, and multiple simplices can be connected along the same face, or even to themselves. We verify the similarities and differences of these complexes via an example of the 2-simplex and an example of the circle S^1 :

Example 4.6 (Simplicial Complex vs. Δ -Complex). **Simplicial Complex: Triangle**

Consider a standard 2-simplex as before, i.e., a filled triangle with vertices v_0, v_1, v_2 . This forms a simplicial complex where:

- The 0-simplices are the vertices: $\{v_0\}, \{v_1\}, \{v_2\}$.
- The 1-simplices are the edges: $[v_0, v_1], [v_1, v_2], [v_0, v_2]$.
- The 2-simplex is the face: $[v_0, v_1, v_2]$.

Every face of a simplex must also be included in the complex, and any two simplices can only meet along shared faces. This satisfies the requirements of a simplicial complex.

Δ -Complex: Circle from One Edge

Now consider a Δ -complex structure on the circle S^1 , defined as follows:

- One 0-simplex v (a single vertex),
- One 1-simplex e , with both endpoints mapped to v ,
- The characteristic map $\sigma : \Delta^1 \rightarrow S^1$ wraps the interval into a loop.

This creates a loop (a 1-dimensional circle) by connecting the endpoints of the 1-simplex at a single vertex. This type of construction isn't allowed in simplicial complexes, because edges must have different vertices, and this example violates that.

This added flexibility makes Δ -complexes more suitable for modelling a wider variety of topological spaces, while still keeping enough combinatorial information to apply methods like simplicial homology.

Next, we define the notion of chain complexes. These provide a structured way to organise information about a space by breaking it down into simpler pieces and describing how these pieces fit together. Chain complexes are fundamental to calculating the homology groups of a discrete group G

4.2 Chain Complexes

Definition 4.7 (*n*th chain group). Let X be a Δ -complex. Then $\Delta_n(X)$ is the free abelian group generated by the characteristic maps $\sigma_\alpha : \Delta^n \rightarrow X$ corresponding to the n -simplices of X .

Elements of $\Delta_n(X)$, called *n-chains*, are formal sums of these maps with integer coefficients, for example,

$$2\sigma_1 - 3\sigma_2 + \sigma_3,$$

where each σ_i is the characteristic map of an n -simplex.

These groups form the building blocks of the chain complex used to define the simplicial homology of X . But to define chain complexes, we must also define the boundary map.

Definition 4.8 (Boundary map). Let X be a Δ -complex, and let $\sigma_\alpha : \Delta^n \rightarrow X$ be the characteristic map of an n -simplex. The boundary map

$$\partial_n : \Delta_n(X) \rightarrow \Delta_{n-1}(X)$$

is defined on the basis elements σ_α by

$$\partial_n(\sigma_\alpha) = \sum_{i=0}^n (-1)^i \sigma_\alpha| [v_0, \dots, \cancel{v_i}, \dots, v_n],$$

where $\cancel{v_i}$ emphasises that v_i gets ignored in each iteration.

Definition 4.9 (Chain complex). Let X be a Δ -complex. The collection of chain groups $\{\Delta_n(X)\}_{n \geq 0}$, together with the boundary maps

$$\partial_n : \Delta_n(X) \rightarrow \Delta_{n-1}(X),$$

forms a *chain complex*

$$\cdots \rightarrow \Delta_{n+1}(X) \xrightarrow{\partial_{n+1}} \Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \rightarrow \cdots$$

such that the composition of two consecutive boundary maps is zero:

$$\partial_n \circ \partial_{n+1} = 0 \quad \text{for all } n.$$

The kernel $\ker(\partial_n)$ are called *cycles*, and the image $\text{Im}(\partial_{n+1})$ are called *boundaries*.

The condition $\partial_n \circ \partial_{n+1} = 0$ means that for any $(n+1)$ -chain, applying the boundary map ∂_{n+1} followed by ∂_n always gives zero. One can think of this as *the boundary of a boundary is zero*.

As a consequence, every element in the image of ∂_{n+1} must lie in the kernel of ∂_n . The kernel of ∂_n consists of all chains whose boundary is zero, which are called *cycles*.

Therefore, we have the inclusion

$$\text{Im}(\partial_{n+1}) \subseteq \ker(\partial_n),$$

which is necessary for the next definition, which takes the quotient of $\ker(\partial_n)$ and $\text{Im}(\partial_{n+1})$.

4.3 Simplicial Homology Groups

Definition 4.10 (*n*th simplicial homology group). Given the chain complex defined above, the *n*th simplicial homology group of X , denoted $H_n(X)$, is defined as the quotient group

$$H_n(X) = \frac{\ker(\partial_n)}{\text{Im}(\partial_{n+1})},$$

where $\ker(\partial_n) \subseteq \Delta_n(X)$ is the subgroup of cycles, and $\text{Im}(\partial_{n+1}) \subseteq \Delta_n(X)$ is the subgroup of boundaries.

Example 4.11 (Homology of the circle S^1). Consider S^1 with the Δ -complex structure consisting of a single vertex v and a single non-degenerate 1-simplex e whose two faces are both v . By Definition 4.7, the simplicial chain groups are

$$\Delta_1(S^1) \cong \mathbb{Z}\langle a \rangle \cong \mathbb{Z}, \quad \Delta_0(S^1) \cong \mathbb{Z}\langle v \rangle \cong \mathbb{Z}, \quad \Delta_n(S^1) = 0 \text{ for } n \geq 2.$$

By Definition 4.8, the boundary is the alternating sum of faces, so

$$\partial_1(e) = v - v = 0,$$

and hence $\partial_1 = 0$; clearly $\partial_n = 0$ for $n \geq 2$ since $\Delta_n(S^1) = 0$, (and by convention, $\partial_0 = 0$).

Thus, the chain complex is

$$\cdots \longrightarrow 0 \longrightarrow \Delta_1(S^1) \xrightarrow{\partial_1=0} \Delta_0(S^1) \longrightarrow 0,$$

so

$$H_0(S^1; \mathbb{Z}) \cong \ker(\partial_0) / \text{im}(\partial_1) \cong \mathbb{Z}/0 \cong \mathbb{Z},$$

$$H_1(S^1; \mathbb{Z}) \cong \ker(\partial_1) / \text{im}(\partial_2) \cong \mathbb{Z}/0 \cong \mathbb{Z},$$

$$\text{and } H_n(S^1; \mathbb{Z}) = 0 \text{ for } n \geq 2.$$

5 Simplicial Cohomology and Cochain Complexes

Having developed simplicial homology, we now turn to its dual theory: simplicial cohomology. While homology is defined via chains of simplices and boundary maps between them, cohomology instead studies functions defined on these simplices. In this way, cohomology offers a different perspective on homology and often yields useful algebraic invariants of a space.

Again, following Section 3.1 in Hatcher's *Algebraic Topology* (Hatcher 2002), we introduce cochain groups as homomorphisms from chain groups into the integers, together with coboundary maps that link these groups into a cochain complex. The cohomology groups are then defined as quotients of cocycles by coboundaries. We show how this construction works with a computation of the cohomology of the circle S^1 .

5.1 Cochain Complexes

Definition 5.1 (*n th cochain group*). Let X be a Δ -complex. The *n th cochain group* of X is defined as

$$C^n(X) = \text{Hom}(\Delta_n(X), \mathbb{Z}),$$

the group of homomorphisms from the n th chain group $\Delta_n(X)$ to the integers $\mathbb{Z}$. Elements of $C^n(X)$, called *n -cochains*, assign an integer to each n -chain in a linear way.

Definition 5.2 (*Coboundary map*). Let X be a Δ -complex and let

$$\varphi \in C^n(X) = \text{Hom}(\Delta_n(X), \mathbb{Z})$$

be an n -cochain. The coboundary map

$$\delta^n : C^n(X) \rightarrow C^{n+1}(X)$$

is defined by

$$(\delta^n \varphi)(\sigma_\beta) = \varphi(\partial_{n+1} \sigma_\beta) = \sum_i (-1)^i \varphi(\sigma_\beta| [v_0, \dots, v_i, \dots, v_{n+1}]),$$

where $\sigma_\beta : \Delta^{n+1} \rightarrow X$ is the characteristic map of an $(n+1)$ -simplex, and ∂_{n+1} is the boundary map on $(n+1)$ -chains.

Note that the coboundary map δ^n takes an n -cochain φ and converts it into an $(n+1)$ -cochain by evaluating φ on the boundary of each $(n+1)$ -simplex.

Now, we have enough to define cochain complexes, in a similar fashion as chain complexes:

Definition 5.3 (*Cochain complex*). Let X be a Δ -complex. The collection of cochain groups $\{C^n(X)\}_{n \geq 0}$, together with the coboundary maps

$$\delta^n : C^n(X) \rightarrow C^{n+1}(X),$$

forms a *cochain complex*. Furthermore, the composition of consecutive coboundary maps is also zero:

$$\delta^{n+1} \circ \delta^n = 0 \quad \text{for all } n \geq 0.$$

Thus, the sequence

$$\dots \xrightarrow{\delta^{n-1}} C^n(X) \xrightarrow{\delta^n} C^{n+1}(X) \xrightarrow{\delta^{n+1}} \dots$$

satisfies $\text{Im}(\delta^n) \subseteq \ker(\delta^{n+1})$ for every n .

5.2 Cohomology Groups

Definition 5.4 (Cohomology group). Let X be a Δ -complex, and let us have a cochain complex as defined above. The n th cohomology group of X is defined as the quotient

$$H^n(X) = \frac{\ker(\delta^n)}{\text{Im}(\delta^{n-1})},$$

where

$$\ker(\delta^n) = \{\varphi \in C^n(X) : \delta^n \varphi = 0\}$$

is the group of *cocycles*, and

$$\text{Im}(\delta^{n-1}) = \{\delta^{n-1} \psi : \psi \in C^{n-1}(X)\}$$

is the group of *coboundaries*.

Example 5.5 (Cohomology of S^1). We compute the cohomology groups of the circle S^1 using a simple Δ -complex structure consisting of:

- one 0-simplex v ,
- one 1-simplex e attached to itself at both ends.

As per the definition, the chain groups are the free abelian groups generated by the n -simplices in the Δ -complex:

$$\Delta_0(S^1) = \mathbb{Z}\langle v \rangle \cong \mathbb{Z}, \quad \Delta_1(S^1) = \mathbb{Z}\langle e \rangle \cong \mathbb{Z},$$

and $\Delta_n(S^1) = 0$ for $n \geq 2$. The boundary map for $n = 1$ yields

$$\partial_1(e) = v - v = 0.$$

The cochain groups are

$$C^0(S^1) = \text{Hom}(\Delta_0(S^1), \mathbb{Z}) \cong \mathbb{Z}, \quad C^1(S^1) = \text{Hom}(\Delta_1(S^1), \mathbb{Z}) \cong \mathbb{Z},$$

and $C^n(S^1) = 0$ for $n \geq 2$.

(Note, the proof of $\text{Hom}(\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}$ is shown after the end of this example.)

The coboundary map $\delta^0 : C^0 \rightarrow C^1$ is given by

$$(\delta^0 \varphi)(e) = \varphi(\partial_1 e) = \varphi(0) = 0,$$

so $\delta^0 = 0$. Since $C^2 = 0$, the map $\delta^1 : C^1 \rightarrow C^2$ is also zero.

We now compute the cohomology groups:

$$H^0(S^1) = \ker \delta^0 \cong C^0(S^1) \cong \mathbb{Z}, \quad H^1(S^1) = \ker \delta^1 / \text{Im} \delta^0 \cong C^1(S^1) / 0 \cong \mathbb{Z},$$

and $H^n(S^1) = 0$ for $n \geq 2$.

Hence, the cohomology groups of the circle are

$$H^n(S^1; \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } n = 0, \\ \mathbb{Z} & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases}$$

□

Remark 5.6. The proof that $\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}$ goes as follows:

Let $\varphi \in \text{Hom}(\mathbb{Z}, \mathbb{Z})$, where we consider $\mathbb{Z}$ as a group under addition. A group homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ is completely determined by the image of $1 \in \mathbb{Z}$, because $\mathbb{Z}$ is a free abelian group on one generator.

Specifically, for any $n \in \mathbb{Z}$, we have

$$\varphi(n) = \varphi(1 + \cdots + 1) = n \cdot \varphi(1)$$

if $n \geq 0$, and for negative n ,

$$\varphi(n) = \varphi(-|n|) = -|n| \cdot \varphi(1) = n \cdot \varphi(1).$$

Thus, every homomorphism φ corresponds uniquely to an integer $k = \varphi(1)$, and we can define an isomorphism

$$\Phi : \text{Hom}(\mathbb{Z}, \mathbb{Z}) \rightarrow \mathbb{Z} \quad \text{by} \quad \Phi(\varphi) = \varphi(1).$$

This map is:

- *Injective:* If $\varphi_1(1) = \varphi_2(1)$, then for all n , $\varphi_1(n) = n \cdot \varphi_1(1) = n \cdot \varphi_2(1) = \varphi_2(n)$, so $\varphi_1 = \varphi_2$.
- *Surjective:* For any $k \in \mathbb{Z}$, define $\varphi_k(n) = kn$. This is a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$, and $\varphi_k(1) = k$, so every integer is in the image.

Therefore, Φ is a group isomorphism, and we conclude that

$$\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}$$

as required.

6 The Universal Coefficient Theorem

Cohomology can be viewed as dual to homology, but the relationship is not always straightforward. The Universal Coefficient Theorem (UCT) provides a precise connection between the two. We will state only the version for cohomology, which is sufficient for our purposes.

To demonstrate how and when the theorem applies, we combine ideas from both Hatcher (2002) and Weibel (1994), and work through explicit examples. While the full algebraic background of the theorem belongs to homological algebra and lies beyond the scope of this project, our aim is to highlight the essential features: when cohomology can be recovered directly from homology, and when additional terms (involving Ext) must be taken into account. In particular, we will see that torsion in homology can contribute directly to cohomology, showing why the UCT is essential for a full understanding.

Theorem 6.1 (Universal Coefficient Theorem for Cohomology, Hatcher 2002, Theorem 3.2). *If a chain complex C of free abelian groups has homology groups $H_n(C)$, then the cohomology groups $H^n(C; G)$ of the cochain complex $\text{Hom}(C_n, G)$ are determined by split exact sequences*

$$0 \longrightarrow \text{Ext}(H_{n-1}(C), G) \longrightarrow H^n(C; G) \xrightarrow{h} \text{Hom}(H_n(C), G) \longrightarrow 0.$$

See Hatcher (2002, Theorem 3.2) for a full proof and further discussion.

Remark 6.2. Formally, Weibel (1994, Def. 2.5.2) defines $\text{Ext}_{\mathbb{Z}}^n(A, \mathbb{Z})$ as a group constructed from a projective resolution of A . But for our purposes, we only require the case where $n = 1$: $\text{Ext}_{\mathbb{Z}}^1(A, \mathbb{Z})$ classifies equivalence classes of short exact sequences

$$0 \longrightarrow \mathbb{Z} \longrightarrow E \longrightarrow A \longrightarrow 0,$$

so it measures the different ways A can arise as an “extension by $\mathbb{Z}$.” In the universal coefficient theorem, this term plays the role of a correction, since cohomology cannot always be recovered directly from homology via $\text{Hom}(A, \mathbb{Z})$. For example, one can check that $\text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0$, so in this case the universal coefficient theorem reduces to an isomorphism

$$H^n(X; \mathbb{Z}) \cong \text{Hom}(H_n(X; \mathbb{Z}), \mathbb{Z}).$$

Example 6.3 (UCT in action for S^1). From earlier examples, we have seen that

$$H_n(S^1; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & n = 0, 1, \\ 0, & n \geq 2, \end{cases} \quad H^n(S^1; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & n = 0, 1, \\ 0, & n \geq 2. \end{cases}$$

Apply Theorem 6.1 with $X = S^1$ and $G = \mathbb{Z}$.

Case $n = 0$: The UCT sequence is

$$0 \longrightarrow \text{Ext}(H_{-1}(S^1), \mathbb{Z}) \longrightarrow H^0(S^1; \mathbb{Z}) \longrightarrow \text{Hom}(H_0(S^1), \mathbb{Z}) \longrightarrow 0.$$

By convention $H_{-1}(S^1) = 0$, so the left term is 0. With $H_0(S^1) \cong \mathbb{Z}$ we get

$$H^0(S^1; \mathbb{Z}) \cong \text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z},$$

exactly as we found before.

Case $n = 1$: The UCT sequence is

$$0 \longrightarrow \text{Ext}(H_0(S^1), \mathbb{Z}) \longrightarrow H^1(S^1; \mathbb{Z}) \longrightarrow \text{Hom}(H_1(S^1), \mathbb{Z}) \longrightarrow 0.$$

Here $H_0(S^1) \cong \mathbb{Z}$ and $H_1(S^1) \cong \mathbb{Z}$, hence $\text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0$ and $\text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}$. Therefore

$$H^1(S^1; \mathbb{Z}) \cong \text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z},$$

as we also found before. If $[e]$ denotes the generator of $H_1(S^1)$, the isomorphism sends a cohomology class of $\varphi \in H^1(S^1; \mathbb{Z})$ to the integer $\varphi([e]) \in \mathbb{Z}$.

Case $n \geq 2$: Since $H_n(S^1) = 0$ for $n \geq 2$, we have $\text{Hom}(H_n(S^1), \mathbb{Z}) = 0$. Also $H_{n-1}(S^1) = 0$ for $n \geq 3$ and $H_1(S^1) \cong \mathbb{Z}$ when $n = 2$, giving $\text{Ext}(H_{n-1}(S^1), \mathbb{Z}) = 0$ in either case anyway. Thus the UCT sequence is $0 \rightarrow 0 \rightarrow H^n(S^1; \mathbb{Z}) \rightarrow 0$, so $H^n(S^1; \mathbb{Z}) = 0$, as computed before.

In all degrees, the UCT isomorphism $H^n(S^1; \mathbb{Z}) \cong \text{Hom}(H_n(S^1), \mathbb{Z})$ reproduces exactly the known (co)homology of S^1 .

A common misconception is that cohomology with integer coefficients always reduces to

$$H^n(X; \mathbb{Z}) \stackrel{?}{\cong} \text{Hom}(H_n(X; \mathbb{Z}), \mathbb{Z}).$$

This is false in general: when $H_{n-1}(X; \mathbb{Z})$ has a property called *torsion*, the Ext term in the Universal Coefficient Theorem (UCT) can contribute non-trivially. So, let us cover an example where the theorem “breaks”.

Remark 6.4 (Torsion). An element $a \in A$ of an abelian group is called a *torsion element* if there exists a non-zero integer m such that $m \cdot a = 0$. The least such positive integer is called the *order* of a . For example, in $\mathbb{Z}/n$ the order of any element divides n , and the generators have order exactly n , so $\mathbb{Z}/n$ is a torsion group. By contrast, $\mathbb{Z}$ has no non-zero torsion elements, so it is torsion-free. More generally, a group may have both torsion and torsion-free parts, for instance $\mathbb{Z} \oplus \mathbb{Z}/2$. See Weibel (1994, Section 3.1) for further discussion, including the relation of torsion to the functors $\text{Tor}_1^{\mathbb{Z}}(A, B)$. While such detail is not needed here, it may provide helpful context.

Example 6.5. A good example is the real projective plane $X = \mathbb{RP}^2$. From Example 2.42 in Hatcher 2002, its homology with integer coefficients is

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}/2, \quad H_2(\mathbb{RP}^2) = 0.$$

Apply Theorem 6.1 (with $G = \mathbb{Z}$):

$$0 \longrightarrow \text{Ext}(H_{n-1}(X), \mathbb{Z}) \longrightarrow H^n(X; \mathbb{Z}) \longrightarrow \text{Hom}(H_n(X), \mathbb{Z}) \longrightarrow 0.$$

Case $n = 1$: Here $H_0(X) \cong \mathbb{Z}$ and $H_1(X) \cong \mathbb{Z}/2$, giving

$$0 \longrightarrow \text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0 \longrightarrow H^1(X; \mathbb{Z}) \longrightarrow \text{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0 \longrightarrow 0,$$

so $H^1(\mathbb{RP}^2; \mathbb{Z}) = 0$. (Here $\text{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0$ because every map from a finite cyclic group to $\mathbb{Z}$ is zero.)

Case $n = 2$: We have $H_1(X) \cong \mathbb{Z}/2$ and $H_2(X) = 0$, so

$$0 \longrightarrow \text{Ext}(\mathbb{Z}/2, \mathbb{Z}) \longrightarrow H^2(X; \mathbb{Z}) \longrightarrow \text{Hom}(0, \mathbb{Z}) = 0 \longrightarrow 0.$$

Thus $H^2(X; \mathbb{Z}) \cong \text{Ext}(\mathbb{Z}/2, \mathbb{Z}) \cong \mathbb{Z}/2$. (The explanation of why $\text{Ext}(\mathbb{Z}/2, \mathbb{Z}) \cong \mathbb{Z}/2$ is true is after this example)

Therefore,

$$H^2(\mathbb{RP}^2; \mathbb{Z}) \cong \mathbb{Z}/2,$$

but,

$$\text{Hom}(H_2(\mathbb{RP}^2), \mathbb{Z}) = \text{Hom}(0, \mathbb{Z}) = 0$$

This example shows that in the UCT, we cannot ignore the Ext term: torsion in $H_{n-1}(X)$ can contribute directly to $H^n(X; \mathbb{Z})$.

Proof. The idea for the proof of $\text{Ext}(\mathbb{Z}/n, \mathbb{Z}) \cong \mathbb{Z}/n$ comes from Weibel (1994, Calculation 3.3.2), and goes as follows:

Apply $\text{Hom}(-, \mathbb{Z})$ to the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n \longrightarrow 0,$$

to obtain the exact sequence

$$0 \rightarrow \text{Hom}(\mathbb{Z}/n, \mathbb{Z}) \rightarrow \text{Hom}(\mathbb{Z}, \mathbb{Z}) \xrightarrow{\times n} \text{Hom}(\mathbb{Z}, \mathbb{Z}) \rightarrow \text{Ext}(\mathbb{Z}/n, \mathbb{Z}) \rightarrow 0,$$

since $\text{Ext}(\mathbb{Z}, \mathbb{Z}) = 0$. We know that

$$\text{Hom}(\mathbb{Z}/n, \mathbb{Z}) = 0 \quad \text{and} \quad \text{Hom}(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z},$$

and the middle map is multiplication by n . Exactness then gives

$$\text{Ext}(\mathbb{Z}/n, \mathbb{Z}) \cong \text{coker}(\times n : \mathbb{Z} \rightarrow \mathbb{Z}) \cong \mathbb{Z}/n.$$

Taking $n = 2$ yields $\text{Ext}(\mathbb{Z}/2, \mathbb{Z}) \cong \mathbb{Z}/2$. □

7 Group homology and the homology of BG

In group homology, a group is studied by assigning to it a sequence of algebraic invariants that reflect structural properties. Classifying spaces, on the other hand, are topological spaces built from groups using the nerve construction. A key result in algebraic topology is that these two viewpoints coincide: the simplicial homology of a group agrees with the (singular) homology of its classifying space BG .

In this section, we show this correspondence in a simplicial setting. First, we show that the nerve of a group carries a natural Δ -complex structure — a fact often stated but rarely proved in detail — which allows us to construct its simplicial chain complex. We then introduce the bar resolution, following Weibel’s Section 6.5 in *An Introduction to Homological Algebra* (Weibel 1994), and prove that it produces the same chain complex.

Theorem 7.1 (Group homology via the classifying space). *Let G be a discrete group. Then, for all $n \geq 0$,*

$$H_n(G; \mathbb{Z}) \cong H_n(BG; \mathbb{Z}).$$

(Corollary 4.1.25 in Löh 2019)

7.1 The realisation of the nerve of a group is a Δ -complex

Let G be a group, viewed as a category as in Example 3.5. Its nerve NG is the simplicial set with

$$(NG)_n = G^n \quad (n \geq 0),$$

with face maps $d_i : G^n \rightarrow G^{n-1}$ and degeneracy maps $s_i : G^n \rightarrow G^{n+1}$ given by

$$\begin{aligned} d_0(g_1, \dots, g_n) &= (g_2, \dots, g_n), \\ d_i(g_1, \dots, g_n) &= (g_1, \dots, g_i g_{i+1}, \dots, g_n) \quad (1 \leq i \leq n-1), \\ d_n(g_1, \dots, g_n) &= (g_1, \dots, g_{n-1}), \\ s_i(g_1, \dots, g_n) &= (g_1, \dots, g_i, e, g_{i+1}, \dots, g_n) \quad (0 \leq i \leq n), \end{aligned}$$

where e is the identity element of G . Its geometric realisation is

$$BG := |NG| = \left(\bigcup_{n \geq 0} \Delta^n \times G^n \right) / \sim,$$

where the equivalence relation is generated by

$$(NG(\phi)(\underline{g}), t) \sim (\underline{g}, \phi_*(t)),$$

for all order-preserving $\phi : [m] \rightarrow [n]$ in Δ , with $\phi_* : \Delta^m \rightarrow \Delta^n$ as the associated affine map.

Lemma 7.2. *A simplex $\underline{g} = (g_1, \dots, g_n) \in G^n$ is degenerate $\iff g_i = e$ for some index $1 \leq i \leq n$.*

Proof. Note that $s_i(g_1, \dots, g_n) = (g_1, \dots, g_i, e, g_{i+1}, \dots, g_n)$. So if any $g_i = e$, we have that $\underline{g}$ is in the image of a degeneracy map, hence $\underline{g}$ is degenerate. Conversely, every degeneracy map s_i inserts an e . $\square$

Now, we can write $(NG)_n^{\text{nd}} := \{(g_1, \dots, g_n) \in G^n : g_i \neq e \forall i\}$ for the non-degenerate simplices.

The following proposition is a specialisation of the standard result that the geometric realisation of a simplicial set carries a natural Δ -complex structure (Hatcher 2002, Chapter 2.1).

Proposition 7.3. *The geometric realisation $BG = |NG|$ has a natural Δ -complex structure with one n -simplex for each non-degenerate tuple $\underline{g} = (g_1, \dots, g_n) \in (NG)_n^{\text{nd}}$. The characteristic map is*

$$\sigma_{\underline{g}} : \Delta^n \longrightarrow BG, \quad \sigma_{\underline{g}}(t) = [t, \underline{g}].$$

Proof. We verify the three axioms of a Δ -complex for the maps $\sigma_{\underline{g}} : \Delta^n \rightarrow BG$, $\sigma_{\underline{g}}(t) = [t, \underline{g}]$, with $\underline{g}$ a non-degenerate n -simplex.

(1): Suppose $\underline{g} \in (NG)_n^{\text{nd}}$. If $\sigma_{\underline{g}}(t) = \sigma_{\underline{g}}(t')$ with $t, t' \in \mathring{\Delta}^n$, then we must have $[t, \underline{g}] = [t', \underline{g}]$ in the realisation. The equivalence relation only identifies t with $\phi_* t$, and $\underline{g}$ with $NG(\phi)(\underline{g})$, via the face and degeneracy maps. Since $\underline{g}$ is non-degenerate, it is not in the image of any degeneracy map. On the other hand, the image of a face map always lies in the boundary of Δ^n , never in its interior. Thus, the only possibility we have is $t = t'$. Hence interior points cannot be identified with any other interior points, so $\sigma_{\underline{g}}$ is injective on $\mathring{\Delta}^n$.

Now let $\underline{g} \in (NG)_n^{\text{nd}}$ and $\underline{h} \in (NG)_m^{\text{nd}}$ be distinct. If $\sigma_{\underline{g}}(t) = \sigma_{\underline{h}}(u)$ with $t \in \mathring{\Delta}^n$, $u \in \mathring{\Delta}^m$, then $[t, \underline{g}] = [u, \underline{h}]$. Again, the equivalence relation only changes the simplex by face or degeneracy maps. But face maps send interiors to boundaries, and degeneracies do not apply to non-degenerate simplices. Thus equality is impossible unless $\underline{g} = \underline{h}$ and $t = u$. So the interiors of distinct non-degenerate simplices are disjoint.

Finally, take any point of BG . It has a representative $[t, \underline{g}]$ with $\underline{g} \in (NG)_n$. If $\underline{g}$ is non-degenerate and $t \in \mathring{\Delta}^n$, we are done. If $\underline{g}$ is degenerate, then the quotient relation shows $[t, \underline{g}] = [t', \underline{h}]$ for some t' in a lower-dimensional simplex and some simplex $\underline{h}$. Iterating this process, every point of BG admits a unique representation $[t', \underline{h}]$ with $\underline{h}$ non-degenerate and $t' \in \mathring{\Delta}^{\dim(\underline{h})}$.

Hence, axiom (1) holds.

(2): Let $\underline{g} \in (NG)_n^{\text{nd}}$. In the quotient relation we have $(\delta^i t, \underline{g}) \sim (t, d_i \underline{g})$ for all $t \in \Delta^{n-1}$. This gives

$$\sigma_{\underline{g}} \circ \delta^i(t) = [\delta^i t, \underline{g}] = [t, d_i \underline{g}] = \sigma_{d_i \underline{g}}(t).$$

If $d_i \underline{g}$ is non-degenerate, then $\sigma_{d_i \underline{g}}$ is the characteristic map of another simplex, so the i th face of $\sigma_{\underline{g}}$ is attached exactly along that map. If $d_i \underline{g}$ is degenerate, then it collapses onto a characteristic map $\sigma_{\underline{h}}$ of some lower-dimensional simplex $\underline{h}$. Thus, each face of $\sigma_{\underline{g}}$ is attached linearly either to another simplex or via a collapse to a lower-dimensional one, as required.

(3): By construction, the realisation BG is given the quotient topology. This means that a subset $U \subseteq BG$ is open if and only if its preimage under every map

$$\Delta^n \times \{\underline{g}\} \longrightarrow BG, \quad (t, \underline{g}) \mapsto [t, \underline{g}],$$

is open in Δ^n . Since every degenerate simplex factors through a non-degenerate one, we can just restrict to the maps

$$\sigma_{\underline{g}} : \Delta^n \rightarrow BG, \quad \underline{g} \in (NG)_n^{\text{nd}}.$$

Thus $U \subseteq BG$ is open if and only if each preimage $\sigma_g^{-1}(U) \subseteq \Delta^n$ is open. This is exactly the condition in the definition of a Δ -complex. Therefore, BG admits a natural Δ -complex structure with simplices indexed by non-degenerate tuples in G^n . $\square$

Remark 7.4 (Faces and degeneracies). The faces of a non-degenerate simplex $(g_1, \dots, g_n)$ in the Δ -complex BG are

$$\begin{aligned} d_0(g_1, \dots, g_n) &= (g_2, \dots, g_n), \\ d_i(g_1, \dots, g_n) &= (g_1, \dots, g_i g_{i+1}, \dots, g_n) \quad (1 \leq i \leq n-1), \\ d_n(g_1, \dots, g_n) &= (g_1, \dots, g_{n-1}). \end{aligned}$$

If a face gains an entry equal to e , it is *degenerate* and factors through a lower-dimensional simplex. Equivalently, it is omitted from the normalised chain group.

Example 7.5. The following two examples of cyclic groups demonstrate how the boundary operator acts in the simplicial chain complex $\Delta_\bullet(BG)$ of the Δ -complex BG :

- Take $G = \langle t \mid t^4 = e \rangle$ and the 2-simplex $(t, t^3) \in \Delta_2(BG)$. Then

$$d_0(t, t^3) = (t^3), \quad d_1(t, t^3) = (tt^3) = (e), \quad d_2(t, t^3) = (t).$$

Hence, the normalised boundary is

$$\partial_2(t, t^3) = d_0 - d_1 + d_2 = (t^3) + (t).$$

- Take $G = \langle t \mid t^3 = e \rangle$ and the 2-simplex $(t, t) \in \Delta_2(BG)$. Then

$$d_0(t, t) = (t), \quad d_1(t, t) = (t^2) \neq e, \quad d_2(t, t) = (t),$$

so no term is dropped and

$$\partial_2(t, t) = (t) - (t^2) + (t).$$

These examples show how the face maps in $\Delta_\bullet(BG)$ either merge adjacent entries or produce the identity, which is then omitted in the normalised chain complex.

7.2 The simplicial chain complex of BG from its Δ -complex structure

Since BG has a natural Δ -complex structure, we can apply Definition 4.7 to define its chain groups.

Recall that for a Δ -complex X , the group $\Delta_n(X)$ is the free abelian group generated by the non-degenerate n -simplices of X , represented by their characteristic maps. In the case $X = BG$, Lemma 7.2 shows that the non-degenerate n -simplices are precisely the tuples

$$(g_1, \dots, g_n) \in G^n \quad \text{with } g_i \neq e \text{ for all } i.$$

Thus the n th chain group of BG is

$$\Delta_n(BG) = \mathbb{Z} \left[\{ (g_1, \dots, g_n) \in G^n : g_i \neq e \ \forall i \} \right].$$

To achieve a chain complex, we now define the boundary maps $\partial_n : \Delta_n(BG) \rightarrow \Delta_{n-1}(BG)$ using Definition 4.8, specialised to the face maps of the nerve NG .

For a generator $(g_1, \dots, g_n) \in \Delta_n(BG)$ with $n \geq 1$, set

$$\partial_n(g_1, \dots, g_n) = (g_2, \dots, g_n) + \sum_{i=1}^{n-1} (-1)^i (g_1, \dots, g_i g_{i+1}, \dots, g_n) + (-1)^n (g_1, \dots, g_{n-1}).$$

We can check that this defines a boundary map ∂_n for our chain complex:

Proof. In the nerve NG , an n -simplex is just a tuple $(g_1, \dots, g_n)$ of group elements. The face maps d_i act by deleting or merging entries:

$$d_0(g_1, \dots, g_n) = (g_2, \dots, g_n), \quad d_i(g_1, \dots, g_n) = (g_1, \dots, g_i g_{i+1}, \dots, g_n), \quad 1 \leq i \leq n-1, \\ d_n(g_1, \dots, g_n) = (g_1, \dots, g_{n-1}).$$

By Definition 4.8, the simplicial boundary is $\partial_n = \sum_{i=0}^n (-1)^i d_i$, which is exactly the formula above. Checking that $\partial_{n-1} \circ \partial_n = 0$ yields

$$\begin{aligned} \partial_{n-1} \circ \partial_n &= \left(\sum_{i=0}^{n-1} (-1)^i d_i \right) \left(\sum_{j=0}^n (-1)^j d_j \right) = \sum_{i=0}^{n-1} \sum_{j=0}^n (-1)^{i+j} d_i d_j \\ &= \sum_{i < j} (-1)^{i+j} d_i d_j + \sum_{i \geq j} (-1)^{i+j} d_i d_j. \end{aligned}$$

Note that

$$\sum_{i \geq j} (-1)^{i+j} d_i d_j = - \sum_{u < v} (-1)^{u+v} d_v d_u.$$

Hence,

$$\begin{aligned} \partial_{n-1} \circ \partial_n &= \sum_{i < j} (-1)^{i+j} d_i d_j - \sum_{i < j} (-1)^{i+j} d_j d_{i+1} \\ &= \sum_{i < j} (-1)^{i+j} (d_i d_j - d_j d_{i+1}) \\ &= 0. \end{aligned}$$

If any entry becomes the identity element, the resulting simplex is degenerate and is removed from the normalised chain group. Hence, ∂_n is a boundary map for BG . $\square$

Therefore, with these maps, the sequence

$$\dots \xrightarrow{\partial_{n+1}} \Delta_n(BG) \xrightarrow{\partial_n} \Delta_{n-1}(BG) \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_2} \Delta_1(BG) \xrightarrow{\partial_1} \mathbb{Z} \xrightarrow{\partial_0} 0$$

is a chain complex.

7.3 The bar resolution

The *bar construction* is a standard way to build a free resolution of a discrete group. We look at the (normalised) bar resolution, following Section 6.5 of Weibel (1994). Let G be a discrete group and $\mathbb{Z}G$ its integral group ring. We regard $\mathbb{Z}$ as a trivial $\mathbb{Z}G$ -module.

Definition 7.6 (Bar resolution). The *bar resolution* is the chain complex of free $\mathbb{Z}G$ -modules

$$\cdots \xrightarrow{d} B_n(G) \xrightarrow{d} B_{n-1}(G) \xrightarrow{d} \cdots \xrightarrow{d} B_1(G) \xrightarrow{d} B_0(G) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0,$$

defined as follows:

- For $n \geq 0$, $B_n(G)$ is the free $\mathbb{Z}G$ -module generated by

$$[g_1, \dots, g_n], \quad g_i \in G, \quad g_i \neq e,$$

i.e. tuples with no entry equal to the identity to avoid degeneracy. Thus a general element of $B_n(G)$ is a finite $\mathbb{Z}$ -linear combination of terms of the form

$$h[g_1, \dots, g_n], \quad h \in G.$$

- The map $\varepsilon : B_0(G) = \mathbb{Z}G \rightarrow \mathbb{Z}$ is given by $\varepsilon([]) = 1$, where $[]$ denotes $1 \in \mathbb{Z}G$.
- The differential $d : B_n(G) \rightarrow B_{n-1}(G)$ is defined on generators by

$$\begin{aligned} d[g_1, \dots, g_n] &= g_1[g_2, \dots, g_n] \\ &\quad + \sum_{i=1}^{n-1} (-1)^i [g_1, \dots, g_i g_{i+1}, \dots, g_n] \\ &\quad + (-1)^n [g_1, \dots, g_{n-1}], \end{aligned}$$

and extended $\mathbb{Z}G$ -linearly.

This complex gives a free $\mathbb{Z}G$ -resolution of $\mathbb{Z}$.

To achieve the n th simplicial homology group of G , i.e. $H_n(G; \mathbb{Z})$, we must take the homology of the chain complex which has chain groups $C_n(G) := \mathbb{Z} \otimes_{\mathbb{Z}G} B_n(G)$.

The tensor product with $\mathbb{Z}$ removes the $\mathbb{Z}G$ -module structure from the complex $B_*(G)$. In $B_*(G)$ each generator is of the form $h[g_1, \dots, g_n]$ with $h \in G$, but in the tensor product, we impose the relation

$$1 \otimes (h \cdot [g_1, \dots, g_n]) = 1 \otimes [g_1, \dots, g_n],$$

since $\mathbb{Z}$ is a trivial right $\mathbb{Z}G$ -module. Thus, every element of $\mathbb{Z} \otimes_{\mathbb{Z}G} B_*(G)$ can be written uniquely as a $\mathbb{Z}$ -linear combination of tensors

$$1 \otimes [g_1, \dots, g_n].$$

In the normalised bar resolution, we ignore the degenerate tuples (those with some $g_i = e$), so the basis consists of the non-degenerate tuples only

$$C_n(G) \cong \mathbb{Z}[\{(g_1, \dots, g_n) \in G^n : g_i \neq e \forall i\}].$$

On generators, the boundary map is

$$\partial(g_1, \dots, g_n) = (g_2, \dots, g_n) + \sum_{i=1}^{n-1} (-1)^i (g_1, \dots, g_i g_{i+1}, \dots, g_n) + (-1)^n (g_1, \dots, g_{n-1}),$$

the same formula as in the simplicial chain complex of BG . Both constructions arise from taking the alternating sum of face maps in the nerve of G so this makes sense.

So, we achieve the following chain complex:

$$\cdots \xrightarrow{\partial_{n+1}} C_n(G) \xrightarrow{\partial_n} C_{n-1}(G) \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0(G) \longrightarrow 0,$$

By definition, the simplicial group homology of G with integer coefficients is the homology of this complex:

$$H_n(G; \mathbb{Z}) := H_n(C_*(G)).$$

7.4 Proof of Theorem 7.1

Proof of Theorem 7.1. Write $\Delta_*(BG)$ for the normalised simplicial chain complex of BG from Section 7.2, and $C_*(G) := \mathbb{Z} \otimes_{\mathbb{Z}G} B_*(G)$ for the (normalised) bar complex from Section 7.3.

Step 1: Chain groups agree on generators. By Lemma 7.2 and Definition 4.7, the group $\Delta_n(BG)$ is the free abelian group on the non-degenerate n -simplices of BG , i.e. tuples $(g_1, \dots, g_n) \in G^n$ with $g_i \neq e$ for all i . Thus,

$$\Delta_n(BG) = \mathbb{Z}[\{(g_1, \dots, g_n) \in G^n : g_i \neq e \forall i\}].$$

On the other hand, tensoring the bar resolution with $\mathbb{Z}$ gets rid of the G -action, so $C_n(G)$ is the free abelian group on the same non-degenerate tuples:

$$C_n(G) \cong \mathbb{Z}[\{(g_1, \dots, g_n) \in G^n : g_i \neq e \forall i\}].$$

Step 2: Boundaries have the same formula. For $n \geq 1$, the boundary map on $\Delta_*(BG)$ is

$$\partial_n(g_1, \dots, g_n) = (g_2, \dots, g_n) + \sum_{i=1}^{n-1} (-1)^i (g_1, \dots, g_i g_{i+1}, \dots, g_n) + (-1)^n (g_1, \dots, g_{n-1}),$$

obtained from the alternating sum of faces; any term that becomes degenerate is removed from the normalised group. After tensoring the bar differential with $\mathbb{Z}$, the boundary on $C_*(G)$ is the same map on the same generators.

Step 3: Define a chain isomorphism. Since both chain groups are free abelian on the same basis of non-degenerate tuples, define

$$\Phi_n : \Delta_n(BG) \longrightarrow C_n(G), \quad \Phi_n(g_1, \dots, g_n) = (g_1, \dots, g_n).$$

To achieve an isomorphism between the chain complexes, we need this commutative diagram to be true:

$$\begin{array}{ccc} \Delta_n(BG) & \xrightarrow{\partial_n} & \Delta_{n-1}(BG) \\ \Phi_n \downarrow & & \downarrow \Phi_{n-1} \\ C_n(G) & \xrightarrow{\partial_n} & C_{n-1}(G) \end{array}$$

that is,

$$\Phi_{n-1} \circ \partial_n^\Delta = \partial_n^C \circ \Phi_n \quad \text{for all } n.$$

By Step 2, the boundary maps agree on generators, so the square commutes. Hence $\Phi_* : \Delta_*(BG) \xrightarrow{\cong} C_*(G)$ is an isomorphism of chain complexes.

Step 4: Take homology. Since $\Phi_* : \Delta_*(BG) \xrightarrow{\cong} C_*(G)$ is an isomorphism of chain complexes, it induces isomorphisms on homology

$$(\Phi_*)_n : H_n(BG; \mathbb{Z}) \xrightarrow{\cong} H_n(C_*(G)).$$

By definition $H_n(C_*(G)) = H_n(G; \mathbb{Z})$, hence

$$H_n(BG; \mathbb{Z}) \cong H_n(G; \mathbb{Z}) \quad \text{for all } n \geq 0,$$

as claimed. □

8 The homology of $B\mathbb{Z}/n$

To show how the homology of a classifying space can be computed, we now consider the case where $G = \mathbb{Z}/n$. Motivated by Chapter 6 of the *Introduction to Homological Algebra* book (Weibel 1994), we compute two complementary approaches: via the Δ -complex structure of $B\mathbb{Z}/n$ (using Proposition 7.3 and Definition 4.9 to obtain the simplicial chain complex), and via the bar resolution. We show that both methods agree, thus providing a worked example of Theorem 7.1.

8.1 Approach 1: Using the lens space model

Theorem 2.10 in Hatcher (Hatcher 2002) tells us that homology groups are homotopy invariant. In particular, Example 1B.4 in Hatcher (Hatcher 2002) says that the classifying space $B\mathbb{Z}/n$ can be modelled as the infinite lens space

$$B\mathbb{Z}/n \simeq S^\infty/(\mathbb{Z}/n),$$

where $\mathbb{Z}/n$ acts freely on $S^\infty \subset \mathbb{C}^\infty$ by scalar multiplication by n th roots of unity:

$$t \cdot (z_0, z_1, \dots) = (\omega z_0, \omega z_1, \dots), \quad \omega = e^{2\pi i/n}.$$

As shown in Example 2.43 of Hatcher's *Algebraic Topology*, $S^\infty/(\mathbb{Z}/n)$ admits a Δ -complex structure with exactly one cell in each dimension that leads to a chain complex with boundary maps alternating between multiplication by 0 and n . Hence, by Definition 4.7, the chain groups are

$$\Delta_k(B\mathbb{Z}/n) \cong \mathbb{Z}, \quad k \geq 0.$$

In addition, these boundary maps come from the way cells of $S^\infty/(\mathbb{Z}/n)$ attach to each other. In odd dimensions, each $(2m+1)$ -cell has two faces that cancel each other because they have opposite orientations, so the boundary is zero. In even dimensions, the $\mathbb{Z}/n$ -action identifies all n faces of a cell with the same lower-dimensional cell, so the boundary map is multiplication by n . This alternating pattern produces the $0, \times n, 0, \times n, \dots$ sequence of differentials in the chain complex.

Hence, we have the following chain complex:

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{0} \mathbb{Z}.$$

Computing homology:

- $H_0 = \ker(\partial_0)/\text{im}(\partial_1) \cong \mathbb{Z}$.
- $H_{2m} = \ker(\partial_{2m})/\text{im}(\partial_{2m+1}) = 0/0 \cong 0$.
- $H_{2m+1} = \ker(\partial_{2m+1})/\text{im}(\partial_{2m+2}) \cong \mathbb{Z}/n$.

Thus, the homology groups are

$$H_i(B\mathbb{Z}/n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & i = 0, \\ \mathbb{Z}/n, & i \text{ odd}, \\ 0, & i \text{ even}, i > 0, \end{cases}$$

exactly as expected but now obtained directly from the topology of the lens space.

8.2 Approach 2: Using the bar resolution

As seen previously, the normalised chain groups $B_k(G)$ in the bar resolution are the free $\mathbb{Z}G$ -modules generated by symbols $[g_1, \dots, g_k]$ with $g_i \neq e$ for all i . Let $G = \mathbb{Z}/n \cong \langle t \mid t^n = e \rangle$. For small k we have

$$\begin{aligned} B_0(\mathbb{Z}/n) &\cong \mathbb{Z}[\mathbb{Z}/n], \quad (\text{generated by } []), \\ B_1(\mathbb{Z}/n) &\text{ free } \mathbb{Z}G \text{ on } [t^a] \text{ with } a \in \{1, \dots, n-1\}, \\ B_2(\mathbb{Z}/n) &\text{ free } \mathbb{Z}G \text{ on } [t^a, t^b] \text{ with } a, b \in \{1, \dots, n-1\}, \\ B_3(\mathbb{Z}/n) &\text{ free } \mathbb{Z}G \text{ on } [t^a, t^b, t^c] \text{ with } a, b, c \in \{1, \dots, n-1\}. \end{aligned}$$

In general,

$$B_k(\mathbb{Z}/n) \cong \bigoplus_{(a_1, \dots, a_k) \in \{1, \dots, n-1\}^k} \mathbb{Z}[\mathbb{Z}/n].$$

To compute homology with trivial coefficients, we form

$$C_k(\mathbb{Z}/n) := \mathbb{Z} \otimes_{\mathbb{Z}[\mathbb{Z}/n]} B_k(\mathbb{Z}/n).$$

Since $\mathbb{Z} \otimes_{\mathbb{Z}[\mathbb{Z}/n]} \mathbb{Z}[\mathbb{Z}/n] \cong \mathbb{Z}$, each summand $\mathbb{Z}[\mathbb{Z}/n]$ in $B_k(\mathbb{Z}/n)$ collapses to a copy of $\mathbb{Z}$. Hence

$$C_k(\mathbb{Z}/n) \cong \mathbb{Z}[\{(t^{a_1}, \dots, t^{a_k}) \mid a_i \in \{1, \dots, n-1\}\}],$$

i.e. the free abelian group on non-degenerate k -tuples (no entry is e).

The boundary maps are the alternating sums of faces; for a generator $(g_1, \dots, g_k)$ with $g_i = t^{a_i}$,

$$\partial_k(g_1, \dots, g_k) = (g_2, \dots, g_k) + \sum_{i=1}^{k-1} (-1)^i (g_1, \dots, g_i g_{i+1}, \dots, g_k) + (-1)^k (g_1, \dots, g_{k-1}),$$

with any term that acquires an e omitted in the normalised complex, and for $\mathbb{Z}/n$, exponents are read mod n . The map $\varepsilon : C_0(\mathbb{Z}/n) \rightarrow \mathbb{Z}$ sends the empty tuple to 1. These maps satisfy $\partial_{k-1} \circ \partial_k = 0$, so

$$\dots \xrightarrow{\partial_3} C_2(\mathbb{Z}/n) \xrightarrow{\partial_2} C_1(\mathbb{Z}/n) \xrightarrow{\partial_1} C_0(\mathbb{Z}/n) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$$

is the required chain complex. From this point, we could, in principle, continue with the bar resolution itself. The computation then becomes increasingly tedious: in even degrees we need to check that every cycle is actually a boundary (so all even homology groups beyond H_0 vanish), while in odd degrees we need to check the cycles have order n . This produces the expected pattern

$$H_{2k}(\mathbb{Z}/n; \mathbb{Z}) = 0 \quad (k \geq 1), \quad H_{2k+1}(\mathbb{Z}/n; \mathbb{Z}) \cong \mathbb{Z}/n.$$

Rather than carrying out these bar computations in detail, a more efficient approach is to use the well-known 2-periodic free resolution of cyclic groups, detailed in Weibel's book (Weibel 1994, Calculation 6.2.1):

$$\dots \xrightarrow{t-1} \mathbb{Z}[\mathbb{Z}/n] \xrightarrow{N} \mathbb{Z}[\mathbb{Z}/n] \xrightarrow{t-1} \mathbb{Z}[\mathbb{Z}/n] \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0, \quad N := 1 + t + \dots + t^{n-1}.$$

Tensoring over $\mathbb{Z}[\mathbb{Z}/n]$ with the trivial module $\mathbb{Z}$ gives

$$(t-1) \otimes 1 = 0 \quad \text{and} \quad N \otimes 1 = (1 + \cdots + 1) = n,$$

so the resulting chain complex is

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0,$$

hence

$$H_i(\mathbb{Z}/n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & i = 0, \\ \mathbb{Z}/n, & i \text{ odd}, \\ 0, & i \text{ even} > 0. \end{cases}$$

8.3 Comparison

Both approaches give identical results, in agreement with Theorem 7.1. In the first approach, we compute the homology groups by identifying the classifying space $B(\mathbb{Z}/n)$ with the infinite lens space $S^\infty/(\mathbb{Z}/n)$, which carries a Δ -complex structure with one non-degenerate simplex in each dimension. Being able to work with homotopic spaces is exactly the benefit of computing via classifying spaces, in most cases choosing a convenient topological model makes the computations much faster and easier to visualise.

By contrast, the bar resolution approach stays entirely within algebra. It is, in principle, universal, but the chain complexes quickly become difficult and tedious to work with. In this case, we were fortunate to have access to Weibel's 2-periodic free resolution, which avoids most of the work while keeping the computation algebraic.

Taken together, the two approaches arrive at the same answer but from different perspectives. The topological model using lens spaces and projections takes a geometric route and makes the calculations more flexible, while the bar resolution stays purely algebraic, though tedious to work with. This example makes it clear how the two methods line up in practice, and it also shows why the classifying space method is usually the more straightforward route, unless a particularly nice resolution is available.

Conclusion

In this dissertation, I have explored the connection between algebra and topology through the study of classifying spaces. Starting with categorical and simplicial foundations, I showed how the nerve of a group leads naturally to a Δ -complex structure on its classifying space BG . This provided a simplicial chain complex whose homology groups can be computed directly.

I also introduced the bar resolution and explained how, after tensoring with the trivial module, it produces the same chain complex. The central theorem, that the homology of a group coincides with the homology of its classifying space, was then established by identifying these two constructions. To show how the theorem worked, I computed the homology of $B\mathbb{Z}$ and $B\mathbb{Z}/n$ by both methods, with complete agreement. These examples show how the topological and algebraic approaches can be equivalent in practice.

This project demonstrates how classical results can be made explicit with relatively accessible tools. There are many natural extensions beyond what I have covered here. A direction one could take is to compute the cohomology rings of classifying spaces, which are a main part to the theory of characteristic classes. Another is to study classifying spaces for more complicated groups, where the geometry of BG contains deeper structural information about the group.

To conclude, classifying spaces shows one of the many ways how algebra and topology are connected. By continually influencing one another, algebra and topology have driven some of the most important progress in mathematics today.

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